

Two-Step Gated GRPO for Long-Horizon Trajectories

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Abstract

We specify a two-step gated GRPO method for long-horizon trajectories. Terminal rewards are averaged within each observed action, distinct action means are standardized equally, and a near-zero first action receives an additional signal only when its successor passes a dispersion gate. We define the resulting advantages and clipped objective; no experiments are reported.

1 Problem and data

Let π_{old} be the known behavior policy, τ_i an old trajectory, and R_i its terminal reward. A state s denotes the complete decision prefix; $a_i(s)$ is the action taken by trajectory i at that state. The aim is to identify first actions with near-zero action-mean advantage whose successor actions show a clearer signed separation.

2 Action-mean advantage at a state

For any observed state s , define

$$\mathcal{I}(s) = \{i : \tau_i \text{ visits } s\}, \quad K_s = |\mathcal{I}(s)|, \quad \mathcal{U}(s) = \{a_i(s) : i \in \mathcal{I}(s)\}. \quad (1)$$

For $u \in \mathcal{U}(s)$, let

$$\mathcal{I}(s, u) = \{i \in \mathcal{I}(s) : a_i(s) = u\}, \quad n_s(u) = |\mathcal{I}(s, u)|, \quad q_s(u) = \frac{1}{n_s(u)} \sum_{i \in \mathcal{I}(s, u)} R_i. \quad (2)$$

Different observed actions then have equal weight, irrespective of their trajectory counts:

$$d_s = |\mathcal{U}(s)|, \quad \bar{q}_s = \frac{1}{d_s} \sum_{u \in \mathcal{U}(s)} q_s(u), \quad \sigma_{q,s} = \left[\frac{1}{d_s} \sum_{u \in \mathcal{U}(s)} (q_s(u) - \bar{q}_s)^2 \right]^{1/2}. \quad (3)$$

The action-mean advantage is returned to every trajectory with that action:

$$A_i^{\text{state}}(s) = \begin{cases} \frac{q_s(a_i(s)) - \bar{q}_s}{\sigma_{q,s}}, & \sigma_{q,s} > 0, \\ 0, & \sigma_{q,s} = 0. \end{cases} \quad (4)$$

The zero-variance rule applies to the advantage at this position, not to the entire trajectory. Repeated actions receive no extra inverse-count weight in the training objective.

3 Two selection rounds and the dispersion gate

At a parent state s_t , use the same action-mean standardization for the first-round advantage and its dispersion:

$$\sigma_t = \sigma_{q,s_t}, \quad A_{1,i}(s_t) = A_i^{\text{state}}(s_t), \quad M_t = \lceil 0.2K_t \rceil. \quad (5)$$

If $\sigma_t = 0$, the state creates no gated candidate. Otherwise, let $\mathcal{L}_t(s_t)$ contain exactly M_t old trajectories with the smallest $|A_{1,i}(s_t)|$. A cutoff tie is filled uniformly without replacement, so only the selected trajectories

retain first-step eligibility. Candidate actions are deduplicated after this trajectory selection:

$$\mathcal{C}_t(s_t) = \{a_{i,t} : i \in \mathcal{L}_t(s_t)\}. \quad (6)$$

For each candidate a , set $s_{t+1} = (s_t, a)$. Use *all* old trajectories reaching this child state, including those not selected in the first round. Require at least five distinct observed next actions:

$$d = |\mathcal{U}(s_{t+1})| \geq 5, \quad \sigma_{t+1} = \sigma_{q, s_{t+1}} > 0. \quad (7)$$

The parent and child dispersions are both equally weighted standard deviations of distinct action-mean rewards. The dispersion gate is

$$G_\Delta(s_t, a) = \mathbf{1}\{\sigma_{t+1} \geq \omega \sigma_t\}, \quad (8)$$

where ω is an unspecified tunable constant. When the gate passes, compute the signed second-step advantage

$$A_2(s_{t+1}, u) = \frac{q_{s_{t+1}}(u) - \bar{q}_{s_{t+1}}}{\sigma_{t+1}}. \quad (9)$$

Let $\mathcal{S}(s_t, a)$ contain the $m(d) = \lceil d/2 \rceil$ distinct actions with largest $|A_2(s_{t+1}, u)|$. Fill cutoff ties uniformly at random. If any child condition or the dispersion gate fails, define $\mathcal{S}(s_t, a) = \emptyset$.

4 Gates and two-step advantages

The action-level gate and trajectory-level first-step gate are

$$G_t^{\text{act}}(s_t, a) = \mathbf{1}\{\sigma_t > 0\} \mathbf{1}\{a \in \mathcal{C}_t(s_t)\} \mathbf{1}\{\mathcal{S}(s_t, a) \neq \emptyset\}, \quad (10)$$

$$B_{i,t} = \mathbf{1}\{i \in \mathcal{L}_t(s_t)\} G_t^{\text{act}}(s_t, a_{i,t}). \quad (11)$$

Thus $B_{i,t} = 1$ implies that at least one successor action passed the second round. A same- a trajectory outside $\mathcal{L}_t$ does not receive the first-step increment

$$\Delta A_{i,t}^{(1)} = \lambda B_{i,t}, \quad \lambda > 0. \quad (12)$$

For a passing child state and a non-overlapping second-step position, define $I_{i,t+1}^{\mathcal{S}} = \mathbf{1}\{a_{i,t+1} \in \mathcal{S}(s_t, a_{i,t})\}$. The second-step coefficient is

$$\tilde{A}_{i,t+1} = [1 + (\beta - 1)I_{i,t+1}^{\mathcal{S}}] A_2(s_{t+1}, a_{i,t+1}), \quad \beta > 1. \quad (13)$$

Since $A_i^{\text{state}}(s_{t+1}) = A_2(s_{t+1}, a_{i,t+1})$ at this child state, unselected actions retain their ordinary action-mean advantage. Selected positive and negative actions both receive a larger absolute coefficient.

If one position is the previous chain's second action and the next chain's selected first action, the confirmed overlap rule adds their two roles:

$$\hat{A}_{i,j}^{\text{overlap}} = A_i^{\text{state}}(s_{i,j}) + \lambda + \beta I_{i,j}^{\mathcal{S}} A_2(s_{i,j}, a_{i,j}). \quad (14)$$

When $I_{i,j}^{\mathcal{S}} = 1$, the action-level part is $(1 + \beta)A_2$, because the two roles are added.

5 Per-position advantage and GRPO update

At each trajectory position, let

$$P_{i,j} = G_{j-1}^{\text{act}}(s_{i,j-1}, a_{i,j-1}), \quad I_{i,j}^{\mathcal{S}} = \mathbf{1}\{a_{i,j} \in \mathcal{S}(s_{i,j-1}, a_{i,j-1})\}. \quad (15)$$

Take a nonexistent previous gate at a trajectory boundary to be zero. Abbreviate $A_{i,j}^{\text{state}} = A_i^{\text{state}}(s_{i,j})$ and $A_{2,i,j} = A_2(s_{i,j}, a_{i,j})$ when $P_{i,j} = 1$. The complete training advantage is

$$\hat{A}_{i,j} = \begin{cases} A_{i,j}^{\text{state}}, & P_{i,j} = 0, B_{i,j} = 0, \\ A_{i,j}^{\text{state}} + \lambda, & P_{i,j} = 0, B_{i,j} = 1, \\ \beta A_{2,i,j}, & P_{i,j} = 1, B_{i,j} = 0, I_{i,j}^S = 1, \\ A_{i,j}^{\text{state}}, & P_{i,j} = 1, B_{i,j} = 0, I_{i,j}^S = 0, \\ \hat{A}_{i,j}^{\text{overlap}}, & P_{i,j} = 1, B_{i,j} = 1. \end{cases} \quad (16)$$

For the known behavior policy π_{old} , define

$$r_{i,j}(\theta) = \frac{\pi_{\theta}(a_{i,j} \mid s_{i,j})}{\pi_{\text{old}}(a_{i,j} \mid s_{i,j})}, \quad (17)$$

$$\ell_{\text{clip}}(r, A) = \min\{rA, \text{clip}(r, 1 - \epsilon, 1 + \epsilon)A\}. \quad (18)$$

Following the per-trajectory and per-position averaging of GRPO [1], maximize

$$J(\theta) = \mathbb{E}_{\{\tau_i\} \sim \pi_{\text{old}}} \left[\frac{1}{N} \sum_{i=1}^N \frac{1}{T_i} \sum_{j=1}^{T_i} \left(\ell_{\text{clip}}(r_{i,j}(\theta), \hat{A}_{i,j}) - \kappa D_{i,j}^{\text{KL}}(\theta) \right) \right]. \quad (19)$$

Equivalently, minimize $-J(\theta)$. Old rewards, gates, and advantages are fixed during an update; gradients pass through the current-policy ratio and an optional KL term. A repeated action contributes one term per old trajectory occurrence, with no additional inverse-count correction.

If KL regularization is enabled, the sampled-action estimator used in [1] can be written

$$D_{i,j}^{\text{KL}}(\theta) = \frac{\pi_{\text{ref}}(a_{i,j} \mid s_{i,j})}{\pi_{\theta}(a_{i,j} \mid s_{i,j})} - \log \frac{\pi_{\text{ref}}(a_{i,j} \mid s_{i,j})}{\pi_{\theta}(a_{i,j} \mid s_{i,j})} - 1. \quad (20)$$

The clip width ϵ , optional KL weight κ , and second-step amplifier β are distinct parameters. Their numerical values are not set here.

6 Calculation order

Collect old trajectories and terminal rewards from the known π_{old} . Group them by complete state and next action, then compute $q_s(u)$, $\sigma_{q,s}$, and $A_i^{\text{state}}(s)$. At each parent state select the lowest 20% of trajectories by $|A_1|$. For each resulting action, use all old trajectories at its child state; check $d \geq 5$, nonzero parent and child dispersions, and the Δ gate. Select $m(d)$ next actions by $|A_2|$, assign $\hat{A}_{i,j}$ using the five cases above, and update θ with the clipped objective.

This document specifies a theoretical procedure, not an experiment. The numerical values of ω , λ , β , and the mapping from an action to a token or full step remain unspecified.

References

- [1] Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. Deepseekmath: Pushing the limits of mathematical reasoning in open language models, 2024. arXiv:2402.03300.